

SOME NOTES ON A-LEVEL MECHANICS

A small discussion on some points in A-level mechanics. Some small understanding-based exercises are provided at the end.

1. Newton's second law of motion: the most important equation in A-level mechanics? Of course, I'm talking about

$$F = ma, \tag{1}$$

understood as “the **resultant** force on an object is equal to its mass times its acceleration”; not to be mixed up with friction ($f \leq \mu R$, see point five).

We apply Newton's second law all the time. If an object is in equilibrium, then $a = 0$, so the resultant force is zero. If an object is moving, we can often determine its acceleration - e.g. with the kinematic (SUVAT) equations - and use this to realise the forces acting on it. Always look to apply $F = ma$ in your problems!

2. Why do we resolve forces? There are two important ideas at play here: (a) that problems in one dimension are much easier to tackle than in higher dimensions, and (b) that forces perpendicular to each other are independent.

By (a), I'm referring to the fact that forces often act on objects in at least two dimensions¹, e.g. with a horizontal (x) and vertical (y) component. Such objects will have a velocity (and hence acceleration) component in both dimensions, and thus it is technically more correct to write (1) as

$$\mathbf{F} = m\mathbf{a} \quad \text{or} \quad \underline{F} = m\underline{a},$$

where the boldface or underlined text indicates a vector-valued quantity.

So far the notion of ‘two dimensions’ has been quite vague. We are most familiar with the x - y axes on a standard graph. What's important is that the horizontal axis is **perpendicular**² to the vertical, and forces that exist purely in one axis have no influence on the other. For example, jumping a perfect vertical would not move you forwards or backwards, and a car's movement along a road does not cause it to take off to the skies.

So long as the axes are perpendicular to each other, they can be constructed as one desires. Problems involving motion on an inclined plane typically rotate the axes so the ‘horizontal’ x component is parallel to the plane, and the y component perpendicular (normal). See Figure 1.

Resolving a force splits it into two components that align with each axis. By doing so, we can consider each axis independently of the other, ‘decoupling’ the dynamics and leading to far simpler analysis.

¹We live in a three dimensional world, but there are many ways to reduce problems down to just two. More importantly, there is great educational merit in considering a simpler case.

²Perpendicular means ‘at a right angle to’ and one may also use ‘orthogonal’ or ‘normal’ (such as the normal force to a slope) to mean the same thing.

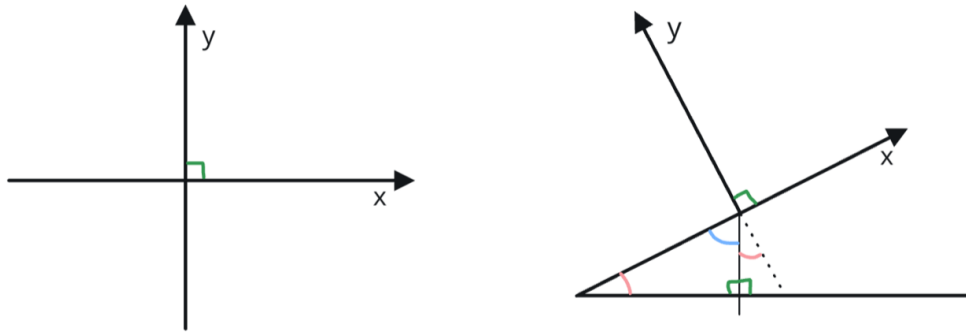


Figure 1: Left: the standard x - y axes. Right: how these axes are rotated on a typical ‘particle on an inclined plane’ problem.

3. Angles on a slope. Refer to Figure 1 again. The angles are colour coded by equality. In particular, the angle between the ground and the inclined plane is equal to the angle between the weight and the rotated y -axis (the extended normal force line).

4. Why is it giving me $\tan(\alpha) = 4/3$ instead of the angle itself? This seemingly annoying formulation of the angle is actually very useful. Remember trigonometry in a right-angled triangle: $\tan(\alpha) = \text{opposite}/\text{adjacent}$. By using Pythagoras’ theorem, you can find the hypotenuse and then use this to quickly find **exact** expressions for $\sin(\alpha)$ and $\cos(\alpha)$, much easier than using α itself.

5. Friction, maximum friction, less than or equal to? Friction, which here we will denote with a lowercase f , has a beautiful relationship with the normal force in $f \leq \mu R$.

The Greek letter μ (‘mew’) is the *coefficient of friction*. One may think of this as describing how rough, or smooth, a surface is³. For example, ice has a far smaller coefficient of friction than carpet.

It is very common to be confused over the inequality $f \leq \mu R$. It can help to consider friction as a ‘lazy force’, in that it only acts on an object if it has to. If a wardrobe sits perfectly stationary (as most wardrobes do) on a carpet, there are no frictional forces at play. However, if you were dragging the wardrobe across the carpet, there would be frictional forces working against you.

This idea can help understand why we have the inequality. If no forces are wanting an object to move, there is nothing for friction to oppose. If a very, very small force acts on an object - like me trying to push a wardrobe across a carpet - a very small frictional force might be enough to counteract the force I’m exerting. However, if an object is in motion, then friction is exerting the maximum force it can.

In sum, friction is at its maximum $f = f_{\max} = \mu R$ if (a) an object is in motion or (b) an object is at a tipping point - ‘any additional force will cause the object to move’. Otherwise, we only have an upper bound $f \leq \mu R$.

³More correctly it measures the interaction between the surface *and* what’s moving across it, but this is good intuition.

Exercise 1.1. *Justify why the angles as described in point three are equal.*

Exercise 1.2. *Draw a straight, diagonal line to resemble a force. Explain, in detail, how we resolve the force into its horizontal and vertical components. You may wish to label the angles between the line and the two axes, and it may be helpful to consider the construction of appropriate triangle(s).*

Exercise 1.3. (a) *Show that if*

$$a^2 + b^2 = c^2,$$

then also

$$(ka)^2 + (kb)^2 = (kc)^2$$

for any real k .

(b) *We call sets of integers (a, b, c) satisfying the Pythagorean equation ‘Pythagorean triples’, with the classical example being $(3, 4, 5)$. Using (a), explain why $(6, 8, 10)$ is a Pythagorean triple. Then, (c) find exact values for $\sin(\alpha)$ and $\cos(\alpha)$ given that*

$$\tan(\alpha) = 5/12.$$

Do not use a calculator. You may check your answers by using the identity $\sin^2(\alpha) + \cos^2(\alpha) = 1$.

Exercise 1.4. *An object sits on a rough, inclined plane. There are no external forces at play, only its weight due to gravity and the normal force to the slope. It sits in equilibrium and is not at the point of slipping down the slope. Explain why friction f is neither zero, nor at its maximum.*